

# Hongsheng Hu / 胡泓昇

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## Basic Information

- **Research interests:** Algebra, representation theory, Lie theory, Coxeter groups.
- **Email:** huhongsheng(at)hnu(dot)edu(dot)cn
- **Homepage:** <https://huhongsheng.github.io>
- **Language:** Chinese (native). English.
- **Birth:** May 16, 1994. Zhangzhou, Fujian Province, China.
- **Other information:** Male. Married. Political status: masses.

## Education and Working Experiences

- Since Jun. 2024. Assistant professor.  
School of Mathematics, Hunan University. Changsha, Hunan Province, China.
- Jul. 2022 – Jun. 2024. Postdoc.  
Beijing International Center for Mathematical Research, Peking University. Beijing, China.  
Mentor: Xiang Fu.
- Feb. 26, 2019 – Aug. 26, 2019. Visiting graduate student.  
University of California, Riverside. California, USA. Host professor: Carl Mautner.  
Funding: China Scholarship Council. No. 201804910559.
- Sep. 2016 – Jun. 2022. Ph.D.  
Academy of Mathematics and Systems Science, Chinese Academy of Sciences. Beijing, China.  
Major in Fundamental Mathematics.  
Supervisor: Nanhua Xi.
- Sep. 2012 – Jun. 2016. B.S.  
Fudan University. Shanghai, China. Major in Mathematics and Applied Mathematics.
- Sep. 2009 – Jun. 2012. Zhangzhou No. 1 High School of Fujian. Zhangzhou, Fujian, China.
- Sep. 2006 – Jun. 2009. Zhao'an Huai'en Middle School. Zhangzhou, Fujian, China.
- Sep. 2000 – Jun. 2006. Zhao'an Experimental Primary School. Zhangzhou, Fujian, China.

## Ph.D. Dissertation (in Chinese)

- Title: Coxeter 群的“反射”表示. (“Reflection” representations of Coxeter groups.)

## Papers and Preprints

- Boundedness of Lusztig's  $\alpha$ -function for Coxeter groups of finite rank.  
j.w. Xiaoyu Chen. arXiv Preprint. 2025.

- Asymptotic log-concavity of dominant lower Bruhat intervals via Brunn–Minkowski inequality.  
j.w. Gaston Burrull and Tao Gui. arXiv preprint. 2024.  
Extended abstract (FPSAC 2024): *Sém. Lothar. Combin.* 91B (2024), Art. 94, 12 pp.
- Strongly dominant weight polytopes are cubes.  
j.w. Gaston Burrull and Tao Gui. arXiv preprint. 2024.
- Representations of Coxeter groups of Lusztig’s  $a$ -function value 1.  
Online, *Comm. Algebra*, 2026.
- On exterior powers of reflection representations, II.  
Online, *Canad. J. Math.*, 2025.
- Weyl group symmetries of the toric variety associated with Weyl chambers.  
j.w. Tao Gui and Minhua Liu. *Bull. Lond. Math. Soc.* 57 (2025), no. 10, 3079–3094.
- Some infinite-dimensional representations of certain Coxeter groups.  
*Ann. Comb.* 29 (2025), no. 1, 101–115.
- 最小二乘解与高斯消元法的注记.  
(A note on least squares solutions and Gaussian eliminations.)  
Teaching paper, in Chinese. *大学数学 (College Mathematics)*, 2024, 40(4):59–62.
- On exterior powers of reflection representations.  
*Bull. Aust. Math. Soc.* 110 (2024), no. 1, 90–102.
- Reflection representations of Coxeter groups and homology of Coxeter graphs.  
*Algebr. Represent. Theory* 27 (2024), no. 1, 961–994.

## Teaching

- Since 2024. Hunan University. Linear Algebra A, Advanced Algebra B.
- 2016 – 2022. University of Chinese Academy of Sciences. Linear Algebra A (teaching assistant), Algebra (teaching assistant), Linear Algebra B (teaching assistant).

## Professional Activities

- Since Jun. 2024. Reviewer for Mathematical Reviews.
- Since Dec. 2021. Reviewer for zbMATH Open.

## Funding

- Jan. 2026 – Dec. 2028. National Natural Science Foundation of China. No. 12501037.
- Jun. 2024 – Jun. 2029. Fundamental Research Funds for the Central Universities.  
No. 531118010972.
- Feb. 2019 – Aug. 2019. China Scholarship Council. No. 201804910559.

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